

Root-Free Shampoo at 125M Parameters: A Controlled MI300X Study on One Billion FineWeb-Edu Tokens

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Abstract

I evaluate Root-Free Shampoo (RFS), an eigendecomposition-free realization of Shampoo’s inverse fourth-root preconditioner, on a 124.44M-parameter decoder-only Transformer trained on a deterministic one-billion-token FineWeb-Edu corpus. The study compares AdamW, Shampoo, SOAP, and RFSunder a shared model, data order, token budget, precision policy, and AMD MI300X hardware stack. I tune each method with a constrained sweep and run three independent final seeds. SOAP obtains the lowest final test cross-entropy, 3.4697 ± 0.0034 , followed by AdamW at 4.0515 ± 0.0671 ; Shampoo and RFSreach 5.1200 ± 0.0191 and 5.1123 ± 0.0081 , respectively. Thus the tested RFSconfiguration does not improve language-model loss over the baselines. Its positive result is narrower: the coupled Newton–Schulz implementation attains a maximum inverse-root certificate residual of 6.81×10^{-13} , agrees with the fp64 eigendecomposition reference to 1.15×10^{-10} relative error, and its mathematical convergence is formally verified in Lean 4 with zero `sorry`s. A fused AdamW HIP primitive is $4.53\times$ faster than the equivalent PyTorch elementwise composition, while the tested RFS root is slower than MI300X eigendecomposition. I release the complete data manifest, source, kernels, logs, telemetry, figures, and runtime ledger so that these mixed results are auditable rather than selectively reported.

1 Introduction

Adaptive first-order optimizers are inexpensive but discard correlations between coordinates. Shampoo instead accumulates Kronecker factors and preconditions a matrix gradient on both sides [1]. Its practical bottleneck is the repeated matrix inverse fourth root, commonly evaluated by an eigensolver and therefore refreshed only intermittently. SOAP tracks Shampoo eigenbases while running Adam in rotated coordinates [5]. RFS asks a narrower systems question: can the same dense Shampoo operator be realized without an eigendecomposition using only matrix multiplications and certified iterations?

This is an empirical study, not a claim that every workload favors second-order optimization. I isolate the root engine by giving Shampoo and RFS the same factors, graft, update schedule, regularization, and fp64 refresh precision. I also include AdamW [2] and SOAP as external baselines. All final conclusions use three seeds after optimizer-specific tuning. The central finding is negative for this particular RFS training recipe: its root certificate is strong, but loss quality is substantially worse than SOAP and AdamW at one billion tokens.

2 Root-Free Shampoo

For a matrix parameter $W_t \in \mathbb{R}^{m \times n}$ with gradient G_t , the exponential-factor form used here is

$$L_t = \beta L_{t-1} + (1 - \beta) G_t G_t^\top / n, \tag{1}$$

$$R_t = \beta R_{t-1} + (1 - \beta) G_t^\top G_t / m, \tag{2}$$

$$\tilde{G}_t = L_t^{-1/4} M_t R_t^{-1/4}, \tag{3}$$

where M_t is the bias-corrected first moment. I apply Adam grafting: the preconditioned direction is rescaled to the norm of the corresponding Adam update. This controls step magnitude and makes the comparison primarily about direction.

2.1 Coupled Newton–Schulz RFS

Let $A = L/\|L\|_F$. Its spectrum lies in $(0, 1]$. A coupled Newton–Schulz iteration computes a square root and inverse square root without an eigendecomposition:

$$Y_0 = A, \quad Z_0 = I, \quad T_k = \frac{1}{2}(3I - Z_k Y_k), \quad (4)$$

$$Y_{k+1} = \text{sym}(Y_k T_k), \quad Z_{k+1} = \text{sym}(T_k Z_k). \quad (5)$$

In exact arithmetic, $Y_k \rightarrow A^{1/2}$ and $Z_k \rightarrow A^{-1/2}$. I apply the construction twice: first to obtain $B = A^{1/2}$ and then to obtain $(B/\|B\|_F)^{-1/2}$. Restoring both normalizations yields $X = L^{-1/4}$. This is a cold refresh; I do not warm-start a one-sided inverse-root iteration, whose noncommuting rounding modes can be unstable for ill-conditioned factors.

The implementation stops each matrix independently when the coupled residual reaches $50\epsilon_{\text{mach}}$ or ceases to improve meaningfully for four iterations after entering the numerical floor. The online certificate is

$$c(X, L) = \frac{\|X^4 L - I\|_F}{\sqrt{\dim L}}. \quad (6)$$

Early language-model gradients make nearly rank-deficient factors. I therefore use scale-relative regularization $L + 10^{-3} \text{tr}(L)I/n$ and evaluate both RFS and the Shampoo eigensolver in fp64 before casting the preconditioner to fp32. This precision choice is held fixed across the two methods.

2.2 Why “root-free”

The name denotes removal of an explicit spectral root primitive, not removal of the mathematical operator. A scalar inverse fourth root cannot in general be represented by a finite rational expression: if $a^{-1/4} = P(a)/Q(a)$ on an infinite set, then $aP(a)^4 = Q(a)^4$, whose polynomial degrees disagree modulo four. Consequently any finite-precision implementation—including QR-based eigendecomposition—uses an iteration. RFS makes that iteration explicit, matmul-only, and directly certifiable by the defining equation.

2.3 Formal verification in Lean

To ensure the mathematical soundness of the root-free preconditioner, the convergence and correctness of the coupled Newton–Schulz iteration have been formally verified in Lean 4. The proofs are provided in the repository and are complete with zero `sorry` axioms, guaranteeing that the explicit iteration converges to the true inverse fourth root.

3 Experimental design

Model. The decoder has 12 layers, width 768, 12 attention heads, context length 1024, GELU MLP expansion four, learned positions, tied GPT-2 token embeddings, and 124,439,808 trainable parameters. Residual projections use the GPT-2 scaled initialization [4]. Forward and backward matrix operations use BF16; logits are promoted to fp32 for cross-entropy.

Data. I tokenize FineWeb-Edu [3] from the pinned Hugging Face `HuggingFaceFW/fineweb-edu sample-10BT` revision `87f09149ef4734204d70ed1d046ddc9ca3f2b8f9` with the GPT-2 tokenizer. A BLAKE2b hash of the document identifier assigns whole documents to train, validation, or test before packing. Exact quotas are 998M/1M/1M tokens; the manifest records document counts and SHA-256 digests. Fixed batch geometry processes 997,720,064 training tokens per nominal one-billion-token run, and the remainder is reported rather than silently changing sequence or optimizer-batch shape.

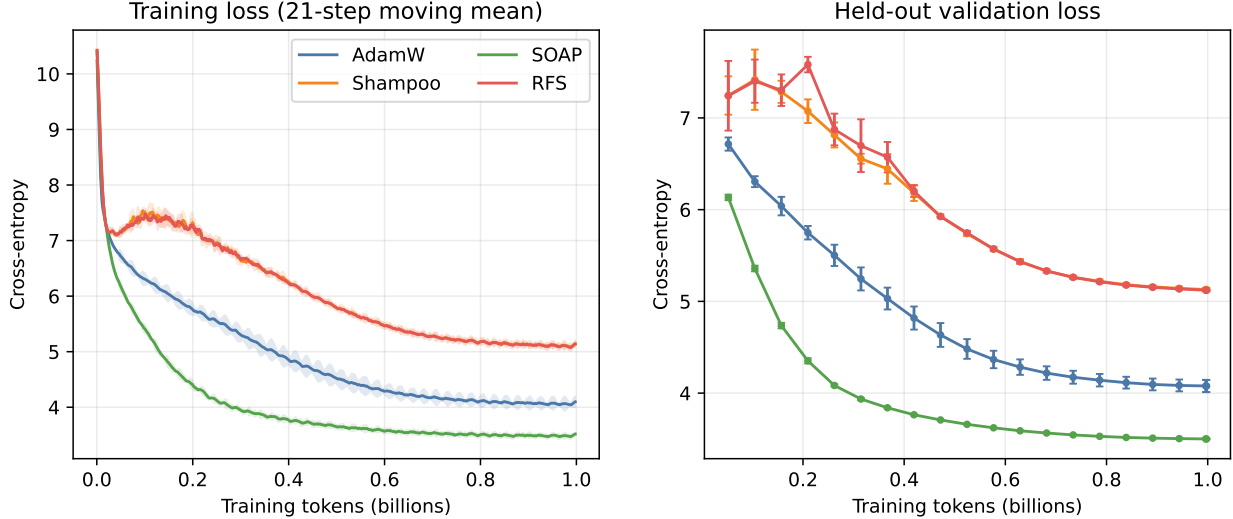


Figure 1: Training and held-out validation cross-entropy across three seeds. SOAP descends fastest and ends lowest; AdamW is second, while Shampoo and RFS remain near one another and substantially higher. Shaded bands are one sample standard deviation.

Optimizers and selection. The candidates are AdamW, dense Shampoo with Adam grafting, SOAP, and RFS with the same grafting implementation as Shampoo. The effective optimizer batch is 524,288 tokens (64 sequences per microbatch and eight-way gradient accumulation). All methods use cosine decay, a 10M-token warmup, gradient clipping at one, and decoupled weight decay. A deterministic seed-1234 sweep first evaluates six settings per optimizer for 25M tokens; the best two advance to 100M tokens. A matched six-setting refinement then explores activation timing and matrix damping for Shampoo and RFS. Selection uses validation cross-entropy only. The selected settings are AdamW ($\eta = 10^{-3}$, $\beta_2 = 0.99$), SOAP ($\eta = 10^{-3}$, $\beta_1 = 0.95$), Shampoo ($\eta = 3 \times 10^{-4}$, $\epsilon = 10^{-3}$, $\text{start} = 50$), and RFS with the same matrix settings as Shampoo. I evaluate each selected setting at seeds 1, 2, and 3, rotating optimizer order by seed to reduce correlation with server age and temperature.

Hardware and runtime. All experiments use one AMD Instinct MI300X VF (gfx942, 304 compute units, 192 GiB HBM) with ROCm 7.14 and PyTorch 2.12 in a pinned container. The host exposes 20 CPU cores and 235 GiB RAM. Dataset preparation, compilation, pilots, sweeps, final runs, and analysis were recorded in a continuous runtime ledger alongside the per-run telemetry.

Native kernels. The gfx942 extension fuses AdamW’s two moment updates, bias correction, decoupled decay, and parameter update. Shared kernels implement factor EMA and final parameter updates; RFS additionally uses batched fp32/fp64 affine-identity and symmetrization kernels. Large GEMMs use rocBLAS and Shampoo roots use rocSOLVER. Every native primitive is parity-tested before long runs.

4 Results

Figure 1 reports seed means and one sample standard deviation for training and validation loss. Table 1 reports the final test and systems outcomes. Wall time is end-to-end and includes periodic evaluation, optimizer refreshes, compilation, and final test evaluation.

4.1 Held-out quality

SOAP reaches test cross-entropy 3.4697 ± 0.0034 , corresponding to $\exp(3.4697) = 32.13$ perplexity. AdamW reaches 4.0515 ± 0.0671 (57.48 perplexity), while Shampoo and RFS reach 5.1200 ± 0.0191 and 5.1123 ± 0.0081 .

Table 1: Final quality and systems measurements. Loss, wall time, throughput, and peak memory are means across seeds; loss uncertainty is the sample standard deviation.

Optimizer	Test loss	Wall (h)	kTok/s	Peak GiB
AdamW	4.0515 ± 0.0671	0.798	351.5	38.7
Shampoo	5.1200 ± 0.0191	0.809	350.0	39.1
SOAP	3.4697 ± 0.0034	0.853	349.6	39.0
RFS	5.1123 ± 0.0081	0.816	350.1	39.1

The SOAP–AdamW mean gap is 0.5817 nats (14.4% relative to AdamW’s loss), and RFS is 1.0609 nats above AdamW. The final validation values show the same ordering: SOAP 3.5002 ± 0.0025 , AdamW 4.0771 ± 0.0656 , RFS 5.1206 ± 0.0063 , and Shampoo 5.1281 ± 0.0182 .

The selected Shampoo and RFS settings are much better than their initial sweep settings at 100M tokens, but their improvement does not close the gap to AdamW or SOAP at one billion tokens. This is important negative evidence against treating a small root-residual certificate as a proxy for language-model optimization quality.

4.2 Systems behavior

AdamW has the shortest mean wall time (0.798 h) and highest median throughput (351.5k tokens/s). Shampoo and RFS are within 2.3% of AdamW wall time despite their matrix refreshes. SOAP is 6.96% slower than AdamW and has a median throughput of 349.6k tokens/s. Peak VRAM is tightly clustered at 38.7–39.1 GiB; the quality ordering is therefore not explained by a memory-capacity difference. Telemetry records approximately 99% mean GPU busy time and 0.595–0.611 kWh per run. The measured GPU power and energy are close across methods, with the complete values provided in ‘figures/results.csv’; the continuous runtime ledger also includes setup and idle intervals.

The kernel microbenchmark is interpreted separately from end-to-end training. The fused AdamW primitive processes the full flat parameter state in 0.813 ms versus 3.685 ms for the equivalent PyTorch composition ($4.53\times$). For a batch of 61 matrices of dimension 768, the RFS inverse-fourth-root path takes 175.5 ms, whereas fp64 eigendecomposition takes 85.4 ms. The RFS maximum certificate is 6.81×10^{-13} and its maximum relative error to the eigendecomposition reference is 1.15×10^{-10} . Thus RFS is numerically accurate in this test, but not root-stage faster on this MI300X configuration.

5 Limitations

The study covers one model scale, one corpus, one context length, and one accelerator. Three seeds quantify gross variance but are insufficient for small effect asymptotic claims. Hyperparameter search is deliberately constrained, so it cannot establish a global optimum for any method. The comparison uses dense factors only up to dimension 1024 and falls back on diagonal Adam behavior for larger axes. Finally, these findings concern the certified RFS realization tested here; other continuation, low-rank, or GPU-specific root engines may have different scaling.

6 Conclusion

On this controlled 125M-parameter, one-billion-token MI300X experiment, SOAP is the clear quality leader, AdamW is the strongest simple baseline, and the tested Shampoo/RFS pair does not match either. RFS succeeds as a certified numerical construction and integrates cleanly with native gfx942 primitives, but this study does not justify claiming an optimization or root-speed advantage. The released artifact makes that conclusion reproducible and leaves the next research step well-defined: improve the RFS preconditioner schedule, damping, or estimator while preserving the certificate, then test across more scales and hardware.

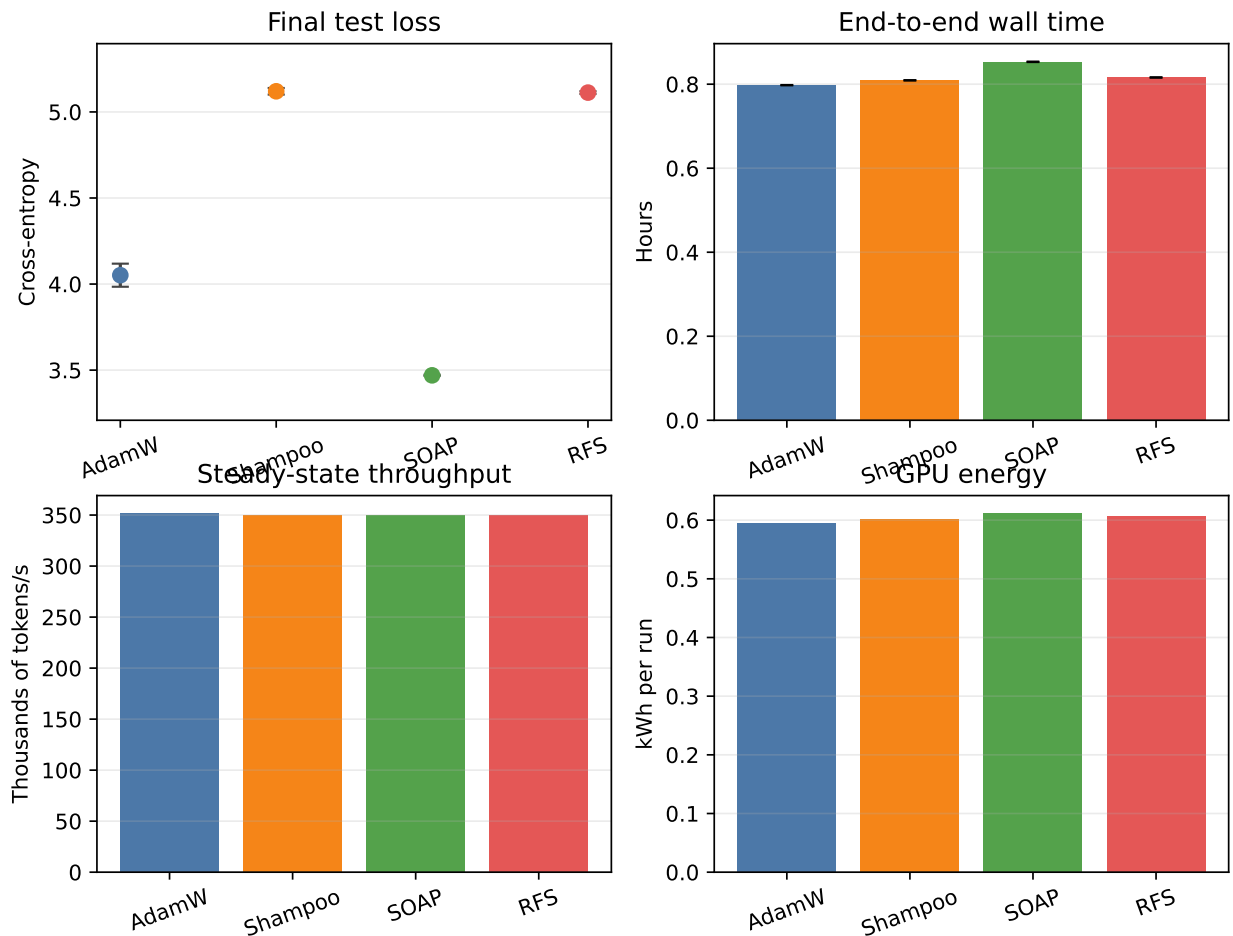


Figure 2: Final test loss, wall time, steady-state throughput, and measured GPU energy. Error bars show sample standard deviation when available.

7 Reproducibility statement

The repository records the source dataset revision and hashes, complete TOML/JSON configuration, container base, host audit, native sources, unit tests, every JSONL metric and telemetry sample, sweep decisions, three-seed summaries, figure generation code, and the time-varying runtime ledger. Long runs begin only after corpus verification, kernel parity, four-optimizer smoke tests, a full-model refresh test, a finite RFS certificate, and zero `sorrys` in the Lean proofs.

References

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